

Literature Benchmark Brief: Wu_2019_IJCAI

Title: Graph WaveNet for deep spatial-temporal graph modeling

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Cluster: Cluster A: Spatio-Temporal Wind/Wake

Core Theoretical Mechanism & Model Formulation

Dilated causal 1D convolution combined with adaptive graph diffusion learning

Empirical Scale & Experimental Dataset

Large-scale traffic sensor networks (METR-LA, PEMS-BAY)

Relevance to IEEE TSTE & Sustainable Energy Systems

Benchmark backbone architecture for spatio-temporal time-series forecasting

Identified Gaps & Operational Vulnerabilities

Lacks aerodynamic inductive biases; symmetric graph diffusion fails to model unidirectional wakes

Role in Manuscript Defense & Reviewer #2 Proof

Benchmark comparator in Table I proving unconstrained STGNNs lack boundary awareness